

Simulator preset and accumulation window comparison

Clip V_DRONE_001 — 640 x 512, 30 fps, 301 frames (10.03 s), one drone labelled throughout. Three simulator presets x three accumulation windows, measured for label alignment.

How alignment is measured

Events are not frames. The simulator produces a continuous stream of (x, y, polarity, timestamp) records with microsecond resolution, while the labels are one box per drone per video frame. To check that a box still lands on its drone after conversion, events have to be gathered into a time window and compared against the box for that frame.

For each video frame, the check gathers every event in the window immediately preceding that frame's timestamp, then computes two quantities:

- **Event share.** The fraction of that window's events falling inside any labelled box.
- **Area share.** The fraction of the image the boxes cover. For this clip the single box covers between roughly 0.46% and 1.19% of the frame (about 1,500 to 3,900 px against 327,680), varying as the drone's apparent size changes. This is a comparatively large target.

The **concentration ratio** is event share divided by area share: how many times denser events are inside the boxes than they would be if scattered at random.

Worked example (clean preset, 20 ms, frame 300): 99.7% of events inside a box covering 0.46% of the frame gives $99.7 / 0.46 = 217x$.

Reading the number

- **~1x** — events inside the boxes are no denser than anywhere else. The boxes are sitting on nothing in particular.
- **2-5x** — weak concentration. Something is there, but it does not stand out from the background.
- **>5x** — clear concentration. The labels are on real structure.
- **>50x** — nearly everything the sensor reports is drone.

What the ratio cannot tell you. It detects gross misplacement — shift every box 50 px and it collapses — but a small consistent offset can still score well, because the drone remains mostly inside a slightly wrong box. It is also depressed by any busy background, even when the labels are perfectly placed, since a textured region adds events outside the boxes and dilutes the share. The rendered images remain the real check; the ratio is a summary that makes 301 frames comparable at a glance.

The three presets

Preset	Cp	Cn	Shot noise	Events (10.0 s)	Rate
clean	0.20	0.20	0 Hz	458,486	45,849 ev/s
default	0.15	0.10	10 Hz	1,685,869	168,587 ev/s
messy	0.05	0.05	15 Hz	2,975,448	297,545 ev/s

Cp and Cn are contrast thresholds in log intensity: how much a pixel's brightness must change before it fires. Lower means a more sensitive sensor. All other simulator parameters were left at their defaults.

The preset names are misleading. "Messy" has *lower* contrast thresholds than default, making it a more sensitive sensor rather than a worse one. Across the three clips measured it produced both more and fewer events than default depending on the scene, so it cannot even be described as the higher-volume preset. A genuinely degraded sensor would be simulated by raising `sigma_threshold` and shot noise while leaving the thresholds at or above default. Names such as "noiseless / baseline / sensitive" would describe what these actually are.

Results: median concentration ratio

Preset	10 ms	20 ms	40 ms	Best
clean	97.8x	98.6x	98.2x	20 ms
default	25.3x	40.3x	35.1x	20 ms
messy	57.9x	75.4x	69.0x	20 ms

Median across all 301 frames, not only the sampled ones.

20 ms wins in every row

Consistent with the other clips measured. The peak is pronounced for the two noisy presets — default gains 60% going from 10 to 20 ms, then loses 13% going to 40 — and essentially absent for the noiseless one.

This is the strongest clip of the three for default

Default scores 40.3x here against 5.5x on V_DRONE_106 and 11.8x on V_DRONE_045 — a seven-fold spread on the same preset. The difference is the target: this clip has a single, comparatively large drone occupying around 1% of the frame, where V_DRONE_106 has two drones at roughly a quarter of that size each. A bigger target produces more events and covers more area, and both help.

The practical reading: default's poor showing elsewhere is not a fixed property of the preset. It fails on small distant targets and copes well with near ones. Any single number quoted for a preset is really a number for a preset on a particular kind of footage.

Per-frame detail at 20 ms

Sampled every 30 frames. "In" is the percentage of that window's events falling inside the box.

Frame	clean ev	clean in	clean x	default ev	default in	default x	messy ev	messy in	messy x
0	0	-	-	0	-	-	0	-	-
30	1,255	97.8%	87.4x	5,745	33.2%	29.6x	7,887	64.0%	57.2x
60	1,127	98.9%	97.7x	4,646	35.1%	34.6x	7,049	63.9%	63.1x
90	1,767	97.5%	91.7x	5,467	41.6%	39.2x	9,414	71.7%	67.4x
120	1,210	99.8%	98.9x	4,279	36.8%	36.4x	6,908	72.1%	71.5x
150	1,204	91.7%	87.9x	5,345	33.6%	32.2x	10,043	48.5%	46.5x
180	579	96.2%	143.0x	4,763	20.3%	30.2x	5,702	44.2%	65.7x
210	1,698	96.6%	81.5x	2,875	70.7%	59.7x	7,899	81.3%	68.6x
240	938	88.2%	124.0x	4,378	24.9%	35.1x	6,880	50.2%	70.6x
270	557	89.0%	153.3x	5,428	10.6%	18.3x	7,595	25.6%	44.0x
300	710	99.7%	216.7x	4,137	17.7%	38.5x	5,075	49.0%	106.5x

What the frames show

Clean: effectively a silhouette

Event share sits between 88% and 99.8% in every sampled frame. At frame 120 it reaches 99.8% — of 1,210 events in that window, all but two came from inside the drone's box. The rendered frame is black except for the drone.

With 458,486 events over ten seconds this is also the densest clean conversion of the three clips — roughly six times V_DRONE_106's and nine times V_DRONE_045's. Sparsity is much less of a concern here; frame 180, the quietest sampled window, still holds 579 events.

Default: usable rather than marginal

Between 10.6% and 70.7% of events inside the box. This is the one clip where the default preset produces data a detector could plausibly work with directly. Frame 210 reaches 70.7% — the drone dominates its window even at 10 Hz shot noise.

Messy: highest event share of the noisy presets

Between 25.6% and 81.3%, roughly double default's at every sampled frame. The pattern holds across all three clips: lower thresholds capture more of the drone's low-contrast edges, and that gain exceeds the cost of the extra shot noise.

Failure modes visible in this clip

Frame 270 is the weakest in both noisy presets

Default drops to 18.3x and messy to 44.0x, both roughly half their clip median, while clean stays at 153.3x. A drop that appears only when noise is present, and not in the noiseless conversion, points to background activity rather than a quiet drone — the drone is still firing normally, but something else in the frame is firing too.

Clean event counts do not track the noisy ones

Frame 210 has the most clean events of any sampled frame (1,698) but the fewest default events (2,875). The two measure different things: clean counts are almost entirely drone, while default counts are dominated by noise and background. A frame being busy in one preset says little about the other.

Frame 0 is empty

The simulator needs a pair of frames to compute a brightness change, so the first frame produces nothing and its window is zero-length. The first labelled frame of every clip has no events behind it.

Caveats on these numbers

- **One clip.** Every figure here comes from V_DRONE_001. It is one of three clips measured; the cross-clip comparison is where general claims belong.
- **Alignment is not detectability.** A high ratio means the labels sit on the events. It does not mean a model can find the drone, and it says nothing about false positives from background structure.
- **Simulated, not measured.** These are synthetic events from RGB video. Interpolation, thresholds and noise models may not match a physical sensor, and no preset here has been validated against real event-camera footage.
- **All conversions were re-run** with a corrected converter after a truncation bug was found in Prophesee's sample script. Any earlier numbers for this clip were measured on incomplete data.